Julian Lozada

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EDUCATION

California State University — **Jan 2025 - Aug 2026**
B.S., Computer Science — *Los Angeles, CA*
- **GPA:** 3.7

PROJECTS

Portfolio Website | Portfolio Website — **Jan 2026 - Feb 2026**
- Built and deployed a production full-stack web application using React, TypeScript, Vite and Java Spring Boot, PostgreSQL, REST API, all serving as a live portfolio with multiple dynamic features
- Engineered responsive UI components with CSS animations, custom hash-based client-side routing, and a component library architecture across 10+ reusable React components
- Architected a full-stack feature pipeline connecting a React/TypeScript frontend to a Spring Boot/PostgreSQL backend, handling data flow from client requests through REST endpoints to database persistence and back

Sea Turtle Tracking System | Research Link — **Sep 2025 - Present**
California State University, Northridge
- Assisted in designing a hardware system integrating sea turtle trackers with machine learning and cloud-based data pipelines to relay real-time migration and ocean condition metrics.
- Developed a web and mobile application to visualize environmental data and turtle movement patters for use by researchers and conservation teams.
- Evaluated and integrated IoUT communication systems to improve underwater data accuracy and reliability.
- Collaborated with the University of Colombia and marine biologists, ensuring technical solutions align with ecological research and conservation goals.
- Produced a technical research paper and project documentation, contributing to the scalability and long-term sustainability of marine tracking initiatives.

TetrisAI | Game Link — **Jan 2026 - Feb 2026**
- Trained a Deep Q-Network (DQN) through a reinforcement learning agent over 10,000+ episodes using PyTorch, implementing experience replay, epsilon-greedy exploration, and custom reward shaping to maximize performance
- Exported the trained neural network to ONNX format and built a live React and TypeScript browser component using ONNX Runtime Web, achieving real-time AI inference at 60fps with no backend dependency
- Delivered an end-to-end ML pipeline spanning game simulation (Python and Pygame), model training, serialization, and client-side web deployment all integrated as a live interactive route on portfolio

EcoSim — **Jan 2026 - Present**
California State University, Northridge
- Co-developed a real-time browser-based ecosystem simulation using React, TypeScript, Three.js, and WebGL2 simulating 1,000+ autonomous agents running entirely on the GPU
- Architected a ping-pong buffer GPU compute system using GLSL shaders, encoding agent state in floating-point textures and sustaining 60fps render performance in the browser without a backend
- Collaborated with a 4-person Agile team to deliver modular GPU simulation engine components, adhering to a shared integration contract that enabled parallel frontend and simulation development across the codebase

PROFESSIONAL EXPERIENCE

Los Angeles Pierce College and California State University, Northridge — **Sep 2022 - Present**
Volunteer Computer Science and Calculus Tutor
- Independently initiated and organized tutoring support outside of official campus resources to help underclassmen with introductory programming, data structures, and algorithm coursework.
- Conducted weekly one-on-one and group sessions focused on breaking down abstract concepts into real-world examples using Java, Python, and C++.
- Built custom study materials, mock exams, and debugging walkthroughs to reinforce problem-solving strategies and build confidence among peers.
- Known for adapting explanations to individual learning styles, resulting in improved student performance and engagement.

SKILLS & INTERESTS

- **Technical**: Python, TypeScript, JavaScript, Java, SQL, HTML 5, CSS3, React, Vite, PostgreSQL, Spring Boot, WebGL2, Jira
- **Interests**: Photography, sports, music, movies, shows